

CAX Scan Procedures

This document is a quick list of procedures needed to efficiently perform mirror, slits and caustic scans in the Carcará-X beamline.

Before the scanning

These are essentially steps to ensure the beam is available and the scripts are working properly.

1. Manually "find the beam"

- Be sure the beamline shutters are open by checking the HMI frontend;
- Open the slit GUI (at the terminal, run: `launch_cax_gui.sh slit1/2`) and check the slits are open;
- Open the DVF1 & 2 GUIs (at the terminal, `launch_cax_gui.sh dvf`); check the 'log' (logarithm) box to help finding the beam;
- Run '`launch_cax_gui.sh mirror`', adjust mirror motor positions; if needed, check last recorded positions before maintenance (archiver: `CAX:A:PB01:m*.RBV`);

2. Run `<path>/cax-scripts/caxmirrorscan-cli` once to check if all PVs are being accessed correctly (the machine state is printed out);

3. (Important) Calibration of the slits span and the DVF image size. The relation between the slit motor span and the DVF1 size is calculated from the YAG dimensions (named `WSIZE_H` and `WSIZE_V`) and motor positions to properly identify the image suitable areas. Check if the hard-coded values converting pixel intervals seen by the DVF to slit motor position are still correct:

- Found in `caxmirrorscan-cli` (functions named `_{top, bottom, left, right}_pos_from_pixels`);
- It can be done by using the DVFA and SlitA GUIs:
 - measure total YAG crystal size in pixels (`WSIZE_V` and `WSIZE_H`);
 - close each blade individually, taking note of initial position and total displacement;
 - update these values in the code.
- Update `Config` class attributes in `<path>/cax-scripts/caxscripts/config.py`.

Obs.: Hopefully, these changes remain permanent or self-adjustable in the near future.

Data acquisition

Once the previous section was accomplished, the scanning should be straightforward.

1. Run `<path>/cax-scripts/caxmirrorscan-cli` and follow the menu directives:

1. Decide what variable to scan (e.g: TX, RY etc);
2. Choose the number of scans (a full scan may be repeated \$n\$ times);
3. Define the initial and final position/angle of a scan;
4. Define how many points to measure in each scan;
5. Choose a suitable file location and name convention: e.g: `mirror_tx_passXX....`

After each scan, the motors should turn back to their initial positions, according to the previously registered machine state.